

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 9: Transport Layer and Queueing

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Deadline: The deadline for this exercise is **23.06.2026, 23:55** and in total **16 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered plagiarism and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Transport Layer Protocols Questions

- (a) Describe how **QUIC** handles connection establishment and security.
- (b) What is **DCCP** and what problem does it aim to solve compared to **UDP** and **TCP**?
- (c) What is the primary purpose of **RTP**, and in which applications is it commonly used?
- (d) What is **SCTP** and what unique features does it offer over **TCP**?
- (e) Explain **SCTP**'s multi-streaming and multi-homing capabilities.

Task 2: Group Exercise: Queueing Questions

- (a) What are the five main aspects of a queueing system that must be defined for analysis, and which performance metrics are typically of interest?
- (b) Explain Little's Law in queueing theory. Under what conditions does it hold, and what does it imply?
- (c) Define the utilization factor (ρ) in the context of a queueing system. Why is it typically required that $\rho < 1$ in stable systems?
- (d) Describe the characteristics of an M/M/1 queue. What are the assumptions for arrival and service processes, and how are the state probabilities computed in steady state?

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- (e) How is the birth-death process related to the M/M/1 queue? Identify the roles of “birth” and “death” in this context.

Task 3: Group Exercise: M/M/1 Queues

A university’s online grading portal runs on a single server. On average, 15 requests arrive every second and it takes the server 50 ms to respond to these requests.

Use an M/M/1-queue model to calculate (you can round to 4 decimal places where necessary):

- (a) the average system time of requests.
- (b) the average utilization of the system.
- (c) the average number of requests that are in the system.
- (d) the probability, that 3 requests are in the system.
- (e) the average idle time in an interval of 8 minutes.

Task 4: Group Exercise: M/M/m Queues (Multiple Servers)

In Utopian City, a new customer service center for computer networking problems with 4 different desks has opened. All counters have an average service time of $x = 2\text{min}$. The customer arrival rate for the respective counters are $\lambda_1 = 0.1 \frac{1}{\text{min}}$, $\lambda_2 = 0.02 \frac{1}{\text{min}}$, $\lambda_3 = 0.03 \frac{1}{\text{min}}$, $\lambda_4 = 0.15 \frac{1}{\text{min}}$. (you can round to 4 decimal places where necessary):

- (a) Calculate the average utilization of the counters.
- (b) Imagine that all representatives are trained to handle inquiries across all services. Calculate the new arrival rate and the average utilization of the system.
- (c) Calculate the probability of the new system being empty.
- (d) Calculate the average system time.
- (e) Calculate the average number of jobs in the system.

Hint: $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$ (for $|x| < 1$)

Task 5: M/M/1 Queues (10 Points)

M/M/1 Queues Theory: Answer the following two questions with the Queuing slides of the lecture!

- (a) State Kendall’s queueing notation and explain the meaning of each component. (1 P)
- (b) Explain the term M/M/1-queue in the context of Kendall’s notation. (1 P)

M/M/1 Queues Practice: Assume that a number of clients request data from a web service which is deployed on a single server machine. On average, 25 requests arrive every second and it takes the server 20 ms to respond to these requests. Use an M/M/1-queue model to calculate (you can round to 4 decimal places where necessary):

- (c) the average system time of requests (2 P)
- (d) the average utilization of the system (1.5 P)
- (e) the average number of requests that are in the system (1.5 P)
- (f) the probability, that 4 requests are in the system (1.5 P)
- (g) the average idle time in an interval of 6 minutes (1.5 P)

Task 6: Congestion Window (6 Points)

In a TCP connection, the size of the congestion window (cwnd) plays a crucial role in determining the transmission rate. The congestion window size can be adjusted based on network feedback, particularly in the following two cases:

- 1. Timeout Event** – Considered a sign of severe congestion, leading to a drastic reduction in the congestion window size.
- 2. Triple Duplicate ACKs** – Treated as a lighter form of congestion, where the window size is reduced moderately.

Suppose a TCP connection initially has a congestion window size of 64 KB, and the Maximum Segment Size (MSS) is 4 KB. The Round-Trip Time (RTT) for this connection is 80 milliseconds.

Assuming the TCP connection follows the Additive Increase Multiplicative Decrease (AIMD) mechanism and begins with an MSS-sized congestion window (i.e., 4 KB), calculate the total time taken for the congestion window to grow back to 64 KB in both the cases (**Timeout Event and Triple Duplicate ACK's**) under the assumption of no further packet losses and ACKs are received regularly (3 Points per Case)